Tharen Candi

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Italian/Australian dual citizen. MSc Computer Science (AI) student working at the intersection of AI and systems: TinyML, embedded systems, computer vision, representation learning. Graduating April 2027, and seeking a graduate role.

Education

University of Padova — Padova, Italy
MSc Computer Science (AI Major) | GPA: 28.08/30 — *Oct 2024 — Apr 2027 (expected)*

University of Sydney — Sydney, AUS
BComp (Computer Science) | WAM: 82.2 (Distinction) — *Feb 2020 — Aug 2023*

EPFL — Lausanne, CH
Exchange Program (Computer Science) | GPA: 5.6/6.0 — *Sep 2022 — Feb 2023*

Research and Projects

MSc Thesis: Closed-Loop On-Device Perception for Tiny Robots, MIG, IAS-Lab — 2026 — Present
- Integrating Rust TinyML inference (microflow-rs) into RIOT OS on ESP32-S3 for closed-loop tiny-robot navigation.
- Writing a RIOT driver for the Arducam OV2640 SPI camera (for upstream) towards a live on-device segmentation demo.
- Developing TinyML models for on-device perception.

The Limits of Cross-Lingual Transfer: SignCLIP on LIS, Research — Feb 2026 — Present
- First study of SignCLIP on low-resource Italian Sign Language. Revealed negative transfer effect for non-iconic signs.
- Ablated six contrastive losses: GlobalNCE raised zero-shot R@1 from 3.7% to 76.2%; linear probe (80.2%) matches SOTA.

Universal Sign Language Representation Learning, Independent Research — 2026 — Present
- Developing a pose-based model that learns cross-lingual sign embeddings by predicting into a frozen text space.

3D Pinna Landmark Extraction, Huawei Munich Tech Arena — Aug 2026 — Sep 2026
- Predicted 85 ear landmarks from 3D scans, fusing DGCNN, HRNet and Point2SSM++ (~1.2 mm mean error).

Multimodal Speech Emotion Recognition, MSc Course Project — 2026
- Late fusion of wav2vec 2.0 audio and MediaPipe face features on RAVDESS.

Hard Real-Time Systems: Ada Ravenscar vs. Rust RTIC, MSc Exam Project — Jan 2026 — Feb 2026
- Compared Ada Ravenscar and Rust RTIC on ARM Cortex-M4, with MAST schedulability analysis and WCET profiling.

Glioblastoma Cell Segmentation Pipeline, Undergraduate Capstone — Feb 2023 — Aug 2023
- Deployed a deep learning pipeline segmenting gigapixel whole-slide images (TensorFlow U-Net, OpenCV, Celery, Redis).

Work Experience

Graduate Intern - Carbon Trading and Origination — Jul 2024 — Sep 2024
Glencore
- Built ETL pipelines for demand forecasting (Python, SQL Server, ActiveBatch), automating a 5-hour manual process.
- Conducted carbon-market modelling and analysis, and wrote technical and end-user documentation.

Private Tutor — Feb 2021 — Jul 2024
Self-employed — *Sydney*
- Tutored university students in OOP and systems programming (C, Java, Python).

Honours and Awards

- **Master's Study Grant**: Italian MFA (2024, 2025)
- **Dean's List**: University of Sydney (2023)
- **Global Mobility Scholarship**: USYD (2022)
- **HSC Top Achiever**: 6th in State for IPT (2019)

Skills

- **Languages**: Python, C/C++, Rust, Ada, Java, Scala
- **ML**: PyTorch, TensorFlow, Hugging Face, OpenCV
- **Systems**: RIOT OS, RTIC, ROS 2, Linux, CUDA, Git
- **Spoken**: English (native), Italian (B2), Greek (A1)